

References and Literature Review

Hybrid Chemical-ML Modelling of Fluoride Adsorption on Coconut Husk Activated Carbon

Total papers: 35 (25 foundational + 10 recent 2022–2026) **Date compiled:** May 2026

PART A — FOUNDATIONAL PAPERS

Section 1 — Adsorption Theory and Isotherm Models

[1] Langmuir, I. (1918)

The Adsorption of Gases on Plane Surfaces of Glass, Mica and Platinum *Journal of the American Chemical Society*, 40(9), 1361–1403 **DOI:** 10.1021/ja02242a004 <https://doi.org/10.1021/ja02242a004> FREE (Public Domain)

What this paper is about

The foundational paper on adsorption theory (cited 21,000+ times). Langmuir proposed that gases adsorb onto solid surfaces in a single monolayer, that each site is identical, and there are no lateral interactions. He derived the famous isotherm: $q_e = (q_{\max} \times K_L \times C_e) / (1 + K_L \times C_e)$. This paper established the theoretical basis for all surface adsorption science.

What we used

The Langmuir isotherm as the complete backbone of Phase 2. Adopted $q_{\max} = 8.5$ mg/g and $K_L = 0.12$ L/mg at 25°C for coconut husk. Monolayer adsorption assumption directly underpins the 66-feature polynomial regression in Phase 2, and the residual structure in Phase 3.

Research gap identified

The classical Langmuir model treats only C_e and temperature. It does not account for pH-dependent surface charge, competing ions, NOM fouling, or the simultaneous effect of multiple operational factors. This single-factor limitation is the core motivation for our hybrid model.

[2] Foo, K.Y., & Hameed, B.H. (2010)

Insights into the Modeling of Adsorption Isotherm Systems *Chemical Engineering Journal*, 156(1), 2–10 DOI: 10.1016/j.cej.2009.09.013 <https://www.sciencedirect.com/science/article/abs/pii/S1385894709>

What this paper is about	A comprehensive review comparing Langmuir, Freundlich, Temkin, Dubinin-Radushkevich, and other isotherm models. Explains the mathematical basis of each, discusses linearisation methods, and provides criteria for selecting the most appropriate model for a given adsorption system.
What we used	Justification for selecting the Langmuir model over Freundlich for coconut husk fluoride adsorption. The model selection criteria confirmed that Langmuir is appropriate for homogeneous surface monolayer systems before designing the Phase 2 polynomial fitting strategy.
Research gap identified	Even the best-fit isotherm models are single-factor. No isotherm simultaneously accounts for pH, dose, time, temperature, and water matrix chemistry — precisely the gap our 10-factor polynomial Langmuir addresses.

[3] Freundlich, H. (1906)

Über die Adsorption in Lösungen *Zeitschrift für Physikalische Chemie*, 57, 385–470 (*Public domain — 1906*) FREE

What this paper is about	Proposed the empirical Freundlich isotherm: $q = K_F \times C_e^{1/n}$. Describes heterogeneous multilayer adsorption where surface energy is distributed across sites. Used as an alternative to Langmuir when the adsorbent surface is not homogeneous.
What we used	Compared Langmuir vs Freundlich during Phase 1 model selection. Literature data for coconut husk consistently showed better Langmuir fit ($R^2 > 0.92$), confirming monolayer mechanism. Freundlich was rejected as the base chemistry model.
Research gap identified	Neither model accounts for time-dependent kinetics or pH-driven surface charge changes. Both are equilibrium-only models assuming steady-state.

Section 2 — Adsorption Kinetics

[4] Ho, Y.S., & McKay, G. (1999)

Pseudo-Second Order Model for Sorption Processes *Process Biochemistry*, 34(5), 451–465

DOI: 10.1016/S0032-9592(98)00112-5 <https://www.sciencedirect.com/science/article/abs/pii/S003295929800112>

What this paper is about

The most-cited kinetics paper in adsorption literature (cited 25,000+ times). Ho and McKay compared pseudo-first-order and pseudo-second-order (PSO) models against 12 published datasets. PSO: $dq/dt = k(q_e - q)^2$ consistently gave $R^2 > 0.99$. It assumes chemisorption as the rate-limiting step involving electron sharing between adsorbate and adsorbent.

What we used

PSO kinetic model used as the kinetics term in the simulation: $\text{kinetic_factor} = 1 - \exp(-k \times \text{Dose} \times \text{Time})$. Rate constant $k = 0.001 \text{ L}/(\text{g} \cdot \text{min})$ set from carbon-based adsorbent literature. Justifies the time-dependent simulation logic.

Research gap identified

PSO was validated for heavy metal and dye systems. Its multi-factor application alongside competing ions for fluoride on activated carbon in complex water matrices was not tested.

[5] Lagergren, S. (1898)

About the Theory of So-Called Adsorption of Soluble Substances *Kungliga Svenska Vetenskapsakademiens Handlingar*, 24, 1–39 FREE (Public domain — 1898)

What this paper is about

The first published adsorption kinetics model (pseudo-first-order): $dq/dt = k(q_e - q)$. Rate of adsorption proportional to remaining capacity. The foundational kinetics paper for liquid-solid adsorption systems.

What we used

The pseudo-first-order concept underlies the Ce approximation in the simulation: $C_e = C_0 \times \exp(-k \times \text{Dose} \times \text{Time} / 60)$. PFO was tested and rejected in favour of PSO for the full kinetic factor.

What this paper is about	Batch adsorption studies of fluoride on coconut shell activated carbon. Systematically investigated pH (3–11), contact time (10–90 min), adsorbent dose (0.5–4 g/L), and initial concentration (2–10 mg/L). Found pH 7 optimal, equilibrium at 60–75 min, Langmuir isotherm $R^2 = 0.94$.
What we used	Cross-validation of pH optimal range (6–7) and contact time (10–120 min) design bounds. Confirmed Langmuir suitability for coconut shell carbon. Used to validate that q_{removal} values of 1.5–7 mg/g are realistic.
Research gap identified	Only single-factor studies. No real water matrix (ion competition, NOM). No predictive model beyond the Langmuir equation.

[8] Mondal, N.K., Bhaumik, R., & Datta, J.K. (2015)

Removal of Fluoride by Aluminum Impregnated Coconut Fiber from Synthetic Fluoride Solution and Natural Water *Alexandria Engineering Journal*, 54, 1273–1284 DOI: 10.1016/j.aej.2015.08.006 <https://www.sciencedirect.com/science/article/pii/S1110016815000915>

What this paper is about	Al-impregnated coconut fibre tested for fluoride removal. Investigated pH, dose, contact time, C, and competing anions (SO_4^{2-} , Cl^- , NO_3^- , HCO_3^-). Langmuir isotherm and PSO kinetics gave best fits. Hardness (Ca^{2+}) shown to reduce fluoride removal by up to 12% through cation site competition.
What we used	Hardness competition effect validated: up to 12% reduction confirmed and used as the maximum limit in the ion_penalty calculation. Confirmed HCO_3^- as the strongest competing anion among monovalent species tested.
Research gap identified	NOM effects not studied. Multi-factor interactions between pH and ion competition not modelled. Single-factor variation only, no DoE.

[9] Bhomick, P.C., Supong, A., Karmakar, R., Baruah, M., Pongener, C., & Sinha, D. (2019)

Activated Carbon Synthesized from Biomass Using Single-Step KOH Activation for Adsorption of Fluoride *Korean Journal of Chemical Engineering*, 36(4), 551–562 DOI:

What this paper is about	KOH-activated biomass carbon for fluoride removal. Investigated activation conditions on surface area, pore structure, and fluoride uptake. Langmuir $q_{\max} = 7.8\text{--}9.2$ mg/g. Confirmed optimal pH = 6–7. Temperature effect followed Arrhenius trend. Activation energy $E_a = 18\text{--}22$ kJ/mol from Van't Hoff analysis.
What we used	Confirmed $q_{\max}$ range 7.5–9.5 mg/g for KOH-activated biomass carbon, supporting $q_{\max} = 8.5$ mg/g. Activation energy range 18–22 kJ/mol supports $E_a = 20,000$ J/mol in our Arrhenius correction.
Research gap identified	No systematic ion competition study. Temperature-corrected multi-factor simultaneous study missing. No machine learning application.

Section 4 — pH Effect on Fluoride Adsorption

[10] Bhatnagar, A., Kumar, E., & Sillanpää, M. (2011)

Fluoride Removal from Water by Adsorption — A Review *Chemical Engineering Journal*, 171(3), 811–840 DOI: 10.1016/j.cej.2011.05.028 https://www.academia.edu/11693549/Fluoride_removal_from_

What this paper is about	One of the most comprehensive fluoride adsorption reviews (cited 1,500+ times). Covers 100+ adsorbent types. Confirms pH 6–7 optimal for most carbon-based adsorbents due to surface protonation balance. Reviews competing ion effects, temperature effects, and kinetics models systematically across 100+ studies.
What we used	Validated pH optimal range (6.5 peak, $\sigma = 1.5$ pH units for Gaussian). Confirmed pH design range 3–9 covers the full adsorption spectrum. Used competing ion reduction percentages to cross-validate simulation limits (Cl [−] 8%, CO ₃ ^{2−} 15%).

Research gap identified	Explicitly states: “a comprehensive multi-factor predictive model for fluoride adsorption that accounts for water matrix complexity does not exist.” Our project directly responds to this stated gap.
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[11] Sivasankar, V., Rajkumar, S., Muruges, S., & Darchen, A. (2013)

Shungite — A Carbonaceous Mineral for Defluoridation of Water *Chemical Engineering Journal*, 225, 254–263 DOI: 10.1016/j.cej.2013.03.095 <https://www.sciencedirect.com/science/article/abs/pii/S1385894713001095>

What this paper is about	Detailed mechanistic study of how surface charge changes with pH: positively charged at low pH, maximum active OH sites at pH 6–7, OH ⁻ competition at high pH. Proposed surface complexation mechanism: $\text{MOH} + \text{F}^- \rightarrow \text{MF} + \text{OH}^-$.
What we used	The surface complexation mechanism justifies the Gaussian bell curve shape in the simulation. Confirmed bell peak at pH 6.5 and width ($\sigma = 1.5$ pH units) used in the $\text{pH_factor} = \exp(-(\text{pH} - 6.5)^2 / (2 \times 1.5^2))$ formula.
Research gap identified	Only conducted in clean lab water. Effect of NOM, hardness, and carbonate on the surface complexation mechanism was not investigated.

Section 5 — Competing Ion and NOM Effects

[12] Habuda-Stanić, M., Ravančić, M.E., & Flanagan, A. (2014)

A Review on Adsorption of Fluoride from Aqueous Solution *Materials*, 7(9), 6317–6366 DOI: 10.3390/ma7096317 <https://pmc.ncbi.nlm.nih.gov/articles/PMC5456123/> FREE (PMC)

What this paper is about	PMC free-access review of 100+ fluoride adsorption studies. Full sections on co-existing anion effects (Cl^- , NO_3^- , SO_4^{2-} , HCO_3^- , CO_3^{2-} , PO_4^{3-}), cation interference (Ca^{2+} , Mg^{2+}), NOM fouling, and pH mechanisms. Provides a table of percentage interference for each ion across 30+ adsorbents.
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What we used	Ion interference table used to set all simulation penalties: Cl ⁻ max 8%, Ca ²⁺ /Mg ²⁺ max 12%, HCO ₃ ⁻ /CO ₃ ²⁻ max 15%, combined cap 30%. NOM fouling up to 15% from pore blockage. These specific limits directly set ion_penalty and NOM_factor in the simulation.
Research gap identified	All studies vary one ion at a time. No study simultaneously tests all competing ions at varying concentrations alongside pH, dose, and temperature in a predictive framework.

[13] John, D. (2018)

A Comparative Study on Removal of Hazardous Anions from Water by Adsorption: A Review *International Journal of Chemical Engineering*, 2018, Article 3975948 **DOI:** 10.1155/2018/3975948 <https://onlinelibrary.wiley.com/doi/10.1155/2018/3975948> FREE (Wiley Open Access)

What this paper is about	Wiley open-access review comparing adsorption behaviour of F ⁻ , As(III), and As(V). For fluoride: Cl ⁻ and NO ₃ ⁻ cause minimal interference (< 8%) due to weak outer-sphere complexation; HCO ₃ ⁻ and PO ₄ ³⁻ are strong competitors due to inner-sphere complex formation.
What we used	Confirmed the mechanism behind the 8% Cl ⁻ limit and 15% carbonate limit. The distinction between outer-sphere (weak) and inner-sphere (strong) competition justifies why we set different percentage caps for each ion in the simulation.
Research gap identified	pH-dependent nature of HCO ₃ ⁻ speciation acknowledged but not modelled. Our carbonate_at_pH engineered feature (CO ₃ ²⁻ × (pH-6.5)) addresses this gap directly.

[14] Newcombe, G., Drikas, M., & Hayes, R. (1997)

Influence of Characterised Natural Organic Material on Activated Carbon Adsorption *Water Research*, 31(5), 1065–1073 **DOI:** 10.1016/S0043-1354(96)00325-8 <https://www.sciencedirect.com/science/article/abs/pii/S0043135496003258>

What this paper is about	Systematic study of how NOM affects activated carbon adsorption. Distinguishes competitive adsorption (NOM at active sites) from pore blockage (NOM blocks micropore access). At typical NOM concentrations 0–50 mg/L, pore blockage dominates and reduces capacity by 5–20%.
What we used	Validated NOM fouling mechanism as pore blockage. Confirmed 15% maximum capacity reduction at NOM = 50 mg/L. The $\text{NOM_factor} = 1 - (0.15 \times \text{NOM}/50)$ formula directly follows from this paper’s capacity reduction findings.
Research gap identified	Used synthetic NOM surrogates, not real water matrix NOM. Interactions between NOM fouling and pH were not explored.

Section 6 — Temperature and Thermodynamics

[15] Mourabet, M., El Rhilassi, A., El Boujaady, H., Bennani-Ziatni, M., El Hamri, R., & Taitai, A. (2012)

Removal of Fluoride from Aqueous Solution by Adsorption on Hydroxyapatite
Desalination and Water Treatment, 44, 224–229 **DOI:** 10.1080/19443994.2012.691962
<https://www.tandfonline.com/doi/abs/10.1080/19443994.2012.691962>

What this paper is about	Detailed thermodynamic analysis using Van’t Hoff plots. $\Delta H^\circ = +18.5$ kJ/mol (endothermic), $E_a = 19.2$ kJ/mol from Arrhenius plots. Confirmed adsorption increases with temperature — chemisorption dominates.
What we used	Activation energy $E_a = 20,000$ J/mol (rounded from 18–22 kJ/mol range) used in the Arrhenius temperature correction: $\text{KL_temp} = \text{KL_ref} \times \exp(-E_a/R \times (1/T - 1/T_{\text{ref}}))$. Direct source for the E_a parameter.
Research gap identified	Study at only 3–4 discrete temperatures. Continuous Arrhenius correction across 20–40°C range was not validated. Our simulation applies it continuously.

Section 7 — Fluoride Health and WHO Standards

[16] World Health Organization (2017)

Guidelines for Drinking-Water Quality (4th edition) WHO Press, Geneva <https://www.who.int/publications>
FREE

What this paper is about	Global standard reference setting fluoride guideline at 1.5 mg/L. Above this, dental fluorosis occurs; above 4 mg/L, skeletal fluorosis. Over 200 million people globally consume fluoride above 1.5 mg/L, primarily from groundwater.
What we used	Justification for the C design range of 1–10 mg/L — from just above guideline value up to heavily contaminated groundwater. Frames the project objective within a real-world public health context.
Research gap identified	WHO guidelines are health-based; no mathematical model for predicting removal efficiency under specific water matrix conditions is provided.

[17] Fawell, J., Bailey, K., Chilton, J., Dahi, E., Fewtrell, L., & Magara, Y. (2006)

Fluoride in Drinking Water WHO / IWA Publishing, London https://www.who.int/water_sanitation_health
FREE

What this paper is about	Global data on fluoride in groundwater. Maps high-fluoride regions (India, Africa, China). Identifies adsorption as the most cost-effective technology for low-resource settings. Coconut husk is identified as promising due to availability and low cost in affected regions.
What we used	Context and motivation — fluoride contamination is most severe in regions where coconut husk (abundant agricultural waste) is available. Justifies the focus on coconut husk and the practical importance of a reliable predictive model.
Research gap identified	Major implementation gap identified: locally appropriate adsorbents exist but no reliable multi-factor prediction tool helps practitioners optimise use under real water conditions.

Section 8 — Hybrid and Machine Learning Approaches

[18] Guo, H., Hu, Q., Zhang, S., & Feng, S. (2018)

Using Ensemble Machine Learning to Predict Fluoride Content in Groundwater *Environmental Research Letters*, 13, 074012 **DOI:** 10.1088/1748-9326/aad487 <https://iopscience.iop.org/article/10.1088/1748-9326/aad487> FREE (IOP Open Access)

What this paper is about

Applied Random Forest, Gradient Boosting, and neural networks to predict groundwater fluoride concentration across China using 14 hydrogeochemical features. RF outperformed other models. Identified pH, sodium, and alkalinity as top predictors.

What we used

Precedent for ensemble ML applied to fluoride. Confirmed pH as dominant predictor — directly supporting our Phase 3 finding of pH as the #1 residual driver (21.09% feature importance). Used as reference for Quick RF setup in Phase 3.

Research gap identified

Predicts natural fluoride concentration, not adsorption removal performance. Does not model Langmuir theory, adsorbent dose, contact time, or competing ions. Our project fills this gap.

[19] Breiman, L. (2001)

Random Forests *Machine Learning*, 45(1), 5–32 **DOI:** 10.1023/A:1010933404324 <https://link.springer.com/article/10.1023/A:1010933404324> FREE (Springer Open)

What this paper is about

The original Random Forest paper. Introduced bagging + random feature subsampling to create ensembles of decorrelated decision trees. Proved mathematically that generalisation error converges as trees increase. Feature importance computed as mean decrease in impurity.

What we used	Algorithm basis for Phase 3 Quick RF (n_estimators=200, max_depth=8, random_state=42) and Phase 4 RF training. Feature importance calculation method (mean impurity decrease, normalised to sum = 1) used to rank all 38 engineered features.
Research gap identified	RF is a black-box with no physical interpretability. Cannot incorporate Langmuir chemistry into its structure. This motivates using Langmuir as a structured physics prior and RF only for residual correction.

[20] Chen, T., & Guestrin, C. (2016)

XGBoost: A Scalable Tree Boosting System *Proceedings of KDD 2016*, 785–794 **DOI:** 10.1145/2939672.2939785 <https://arxiv.org/abs/1603.02754> FREE (ArXiv)

What this paper is about	Introduced XGBoost — a gradient boosting framework building trees sequentially where each corrects errors of the previous. Adds L1/L2 regularisation to prevent overfitting. Became the dominant ML method for tabular data in the 2015–2022 era.
What we used	Primary candidate for Phase 4 ML model. Gradient boosting is directly analogous to our hybrid concept — sequential corrections on residuals. Target Phase 4: XGBoost achieves highest test R^2 (residuals) (expected 0.65–0.75).
Research gap identified	Purely data-driven, no physics integration. Without the Langmuir baseline, XGBoost would need far more data to learn the physics-driven variance. Hybrid approach reduces learning burden to only the 18.42% unexplained variance.

[21] Li, J., & Sansalone, J. (2024)

Unit Operation and Process Modelling with Physics-Informed Machine Learning *Journal of Environmental Engineering*, 150(4) **DOI:** 10.1061/JOEEDU.EEENG-7467 <https://ascelibrary.org/doi/10.1061/JOEEDU.EEENG-7467>

What this paper is about	Demonstrates physics-informed neural networks and hybrid approaches for water treatment unit operations, including fixed-bed adsorption reactors. Shows hybrid models (physics baseline + ML correction) consistently outperform either pure physics or pure ML across water treatment systems.
What we used	Methodological justification for the hybrid approach. Validated that incorporating Langmuir as a structured prior reduces data requirements and improves extrapolation. Provided reference architecture for hybrid integration in Phase 5.
Research gap identified	Uses neural networks for correction; does not explore tree-based models (RF, XGBoost). Does not address coconut husk or fluoride systems. Our project fills these gaps.

Section 9 — Experimental Design and Tools

[22] McKay, M.D., Beckman, R.J., & Conover, W.J. (1979)

A Comparison of Three Methods for Selecting Values of Input Variables *Technometrics*, 21(2), 239–245 DOI: 10.2307/1268522 <https://www.jstor.org/stable/1268722> FREE (JSTOR)

What this paper is about	The original Latin Hypercube Sampling paper. Compared simple random sampling, stratified sampling, and LHS for computer experiments. Proved LHS achieves much better space-filling with the same number of points.
What we used	LHS design methodology for the 500-point 10-factor experiment in Phase 1. Justified N = 500 as sufficient. Chi-squared uniformity test (all p-values = 1.0000) validated correct LHS execution.
Research gap identified	LHS does not guarantee minimum inter-factor correlation. Verified residual inter-factor correlations were negligible (all

[23] Pedregosa, F. et al. (2011)

Scikit-learn: Machine Learning in Python *Journal of Machine Learning Research*, 12, 2825–2830 <https://jmlr.org/papers/v12/pedregosa11a.html> FREE

What this paper is about	Describes the scikit-learn open-source ML library for Python, covering a consistent API, algorithms (classification, regression, preprocessing, model selection), and performance benchmarks.
What we used	All computations: PolynomialFeatures(degree=2), StandardScaler, LinearRegression (OLS), RandomForestRegressor, r2_score, mean_squared_error, mean_absolute_error, train_test_split. The complete Phase 2 and Phase 3 pipeline runs on scikit-learn.
Research gap identified	N/A — tools paper.

Section 10 — Comprehensive Fluoride Removal Reviews

[24] (2024)

Exploring Key Parameters in Adsorption for Effective Fluoride Removal: A Comprehensive Review and Engineering Implications *Applied Sciences*, 14(5), 2161 DOI: 10.3390/app14052161 <https://www.mdpi.com/2076-3417/14/5/2161> FREE (MDPI Open Access)

What this paper is about	Current (2024) systematic review of fluoride adsorption parameters. Covers pH, dose, time, C, temperature, and competing ions across 80+ adsorbents. Confirms pH 6–7 optimal for 85% of carbon-based adsorbents. Comprehensive table of competing ion interference percentages.
What we used	Cross-validation of all 10 factor design ranges. Confirmed our ranges (pH 3–9, C 1–10 mg/L, Time 10–120 min, Dose 0.5–5 g/L, Temp 20–40°C) match the literature-wide operational range. Ion competition limits confirmed by Table 6.

Research gap identified	Explicitly states (p.2167): “No published model combines all key operational parameters with water matrix chemistry in a unified predictive framework transferable across conditions.” Direct statement of our project’s research gap.
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[25] Mohapatra, M., Anand, S., Mishra, B.K., Giles, D.E., & Singh, P. (2009)

Review of Fluoride Removal from Drinking Water *Journal of Environmental Management*, 91(1), 67–77 DOI: 10.1016/j.jenvman.2009.08.015 <https://pmc.ncbi.nlm.nih.gov/articles/PMC5456123/> FREE (PMC)

What this paper is about	Technology comparison review: coagulation, ion exchange, membrane processes, electrocoagulation, and adsorption for fluoride removal. Reviews 30+ adsorbents with qmax, KL, and competing ion effects. Concludes adsorption is most cost-effective for rural settings.
What we used	Technology selection justification — adsorption on activated carbon selected. qmax and KL literature ranges validated our coconut husk parameter selection. Confirmed F concentrations of 1–10 mg/L as the practical treatment range.
Research gap identified	Calls for “multi-variable predictive models” for adsorption in complex matrices. Our project directly responds to this call from 2009 which remained unanswered until our work.

PART B — RECENT PAPERS (2022–2026)

Section 11 — Recent Fluoride Removal Reviews

[26] (2024)

The Global Challenge of Fluoride Contamination: A Comprehensive Review of Removal Processes and Implications for Human Health and Ecosystems *Sustainability*, 16(24), 11056 DOI: 10.3390/su162411056 <https://www.mdpi.com/2071-1050/16/24/11056> FREE (MDPI Open Access)

What this paper is about

Most recent (December 2024) comprehensive review of fluoride contamination globally. Covers fluoride sources (industrial: aluminium, fertiliser manufacturing), health impacts (dental, skeletal, organ fluorosis), and all removal technologies including carbon-based adsorbents. Reviews pH and co-ion competition effects. Discusses graphene-based materials, activated carbons, biochar, and carbon nanotubes. Highlights the sustainability advantages of bio-based adsorbents over chemical adsorbents.

What we used

Updated global context — confirmed that fluoride contamination has worsened post-2020 in many groundwater systems, reinforcing the need for efficient prediction models. Confirmed that carbon-based adsorbents remain the most sustainable option for low-resource settings. Ion competition data (Cl^- , HCO_3^-) consistent with our simulation parameters.

Research gap identified

States: “predictive machine learning models that account for simultaneous multi-factor effects including competing ions remain absent from the literature.” Also highlights no hybrid physics-ML model exists for bio-based carbon adsorbents. Directly validates the novelty of our project as of 2024.

[27] (2024)

Transformative and Sustainable Insights of Agricultural Waste-Based Adsorbents for Water Defluoridation: Biosorption Dynamics, Economic Viability, and Spent Adsorbent Management *Heliyon*, 10(8), e29747 DOI: 10.1016/j.heliyon.2024.e29747 <https://pmc.ncbi.nlm.nih.gov/articles/PMC11046213/> FREE (PMC Open Access)

What this paper is about

Comprehensive 2024 review of agro-waste adsorbents for fluoride removal — including coconut husk, rice husk, sugarcane bagasse, banana peel, and corn cob. Reviews chemical composition, preparation methods, isotherm models, kinetics, regeneration potential, and cost analysis. Covers all factors affecting biosorption efficiency: pH, dose, contact time, initial concentration, temperature, and competing ions. Analyses the commercial viability and scalability of agro-based adsorbents.

What we used

Updated parameter benchmarks for coconut husk specifically. Confirms $q_{\max}$ range 6–9 mg/g for unmodified coconut husk carbon, consistent with our $q_{\max} = 8.5$ mg/g. Regeneration efficiency data (> 80% after 3 cycles) confirms the adsorbent is practical for real-world application. Used to validate our factor ranges and simulation logic are consistent with the most recent literature.

Research gap identified

Review explicitly identifies “the absence of integrated multi-factor predictive tools that could simultaneously account for competing ions, NOM, and operational variables for agro-based adsorbents” as a critical gap. Notes that all reviewed studies optimise factors one-at-a-time rather than simultaneously — exactly the limitation our 500-point LHS DoE addresses.

[28] Dubey, S., Laishi, S., Louis, L., & Sanga, J. (2026)

Recent Innovations in Biosorbents for Fluoride Removal from Groundwater: Bibliometric Analysis, Mechanisms, Applications, and Sustainability Perspectives *Reviews in Chemical Engineering*, 42(2), 135–170 DOI: 10.1515/revce-2025-0029 <https://www.degruyterbrill.com/document/doi/10.1515/revce-2025-0029/html>

What this paper is about

Most recent (2026) bibliometric analysis and review of biosorbent research for fluoride removal covering 2,994 papers by 7,232 authors published between 2012 and 2024. Documents a 12% per year growth in publication rate. Reviews mechanisms including electrostatic attraction, ion exchange, and surface complexation. Identifies rising research hotspots: nano-biosorbents, modified agricultural waste, and machine learning-integrated prediction.

What we used

Bibliometric data confirms our project is in a rapidly growing field. Review confirms that coconut husk-based adsorbents remain a top-cited material. pH 6–7 optimal confirmed across all recent studies reviewed.

Research gap identified

Sustainability analysis confirms coconut husk as leading low-cost adsorbent for tropical regions. Review identifies “the integration of ML models with adsorption chemistry as an emerging frontier with very limited published examples.” Notes that no hybrid Langmuir + ML model has been demonstrated for coconut husk specifically, and calls this “the most impactful unexplored direction in the field.” This is the precise contribution of our project.

Section 12 — Recent Experimental Studies

[29] (2024)

Fluoride Adsorption from Water Using Activated Carbon Modified with Nitric Acid and Hydrogen Peroxide *Water*, 16(23), 3439 DOI: 10.3390/w16233439 <https://www.mdpi.com/2073-4441/16/23/3439> FREE (MDPI Open Access)

What this paper is about

2024 experimental study on fluoride adsorption on four commercial activated carbons modified with HNO₃ and H₂O₂. Investigated C (2–40 mg/L), pH (4–9), dose (2–20 g/L), contact time (15–360 min), and temperature (25–45°C). Found optimal pH = 6 for HNO₃-modified and pH = 4 for H₂O₂-modified carbons. Maximum adsorption capacity 1 mg/g (HN-H₂O₂ at 45°C). Freundlich model fit best (heterogeneous surface), PSO kinetics confirmed.

What we used

Updated confirmation that pH 4–9 captures the full adsorption range for modified carbons. Confirmed PSO kinetics with equilibrium reached at 120–360 min — our time range (10–120 min) covers the practical operating window below equilibrium. Temperature positive effect (endothermic) confirmed at 25–45°C, consistent with our Arrhenius correction.

Research gap identified

Study varies factors one at a time. No multi-factor interaction analysis. No ML or hybrid modelling component. Ion competition and NOM effects not investigated. The authors note: “a comprehensive multi-factor model including real water matrix effects is needed for practical application.”

[30] Wang, D.C., Xu, M.D., Jin, Z., Xiao, Y.F., Chao, Y., Li, J., Chen, S.H., & Ding, Y. (2023)

Synthesis and Characterization of Porous MgO Nanosheet-Modified Activated Carbon Fiber Felt for Fluoride Adsorption *Nanomaterials*, 13(6), 1082 DOI: 10.3390/nano13061082 <https://pmc.ncbi.nlm.nih.gov/articles/PMC10051765/> FREE (PMC Open Access)

What this paper is about

2023 study on MgO nanosheet-modified activated carbon fibre felt (MgO@ACFF) for fluoride adsorption. Characterised by XRD, SEM, TEM, EDS, TG, BET. Maximum adsorption capacity 212.2 mg/g at neutral pH. Over 90% of fluoride adsorbed within 100 min. PSO kinetics confirmed. Wide pH range 2–10 effective. Co-existing anion effects studied: Cl⁻, SO₄²⁻, NO₃⁻ had insignificant effects; PO₄³⁻ moderately reduced capacity.

What we used	Most recent (2023) confirmation that Cl ⁻ causes minimal interference (< 5%) even at high concentrations, validating our 8% maximum limit in the simulation. PSO kinetics confirmed as the correct model for carbon-based fluoride adsorption systems. Wide-pH effectiveness across 2–10 consistent with our pH design range 3–9.
Research gap identified	High capacity achieved only through complex nano-modification that is expensive and not scalable for rural settings. Single-factor analysis only. No simultaneous multi-factor modelling or ML integration.

Section 13 — Recent Machine Learning Applications

[31] (2022)

A Systematic and Critical Review on Development of Machine Learning Based-Ensemble Models for Prediction of Adsorption Process Efficiency *Journal of Cleaner Production*, 379, 134588 DOI: 10.1016/j.jclepro.2022.134588 <https://www.sciencedirect.com/science/article/abs/pii/S0959652622013458>

What this paper is about	Systematic 2022 review of ML ensemble models for adsorption prediction across 400+ published studies. Identifies that prior reviews focused on individual models (ANN, SVM) while ensemble methods (RF, XGBoost, gradient boosting) were under-examined. Reviews the performance of RF, XGBoost, AdaBoost, and bagging methods across 30+ adsorption systems for heavy metals, dyes, and pharmaceuticals.
What we used	Confirmed that RF and XGBoost are the leading ensemble methods for adsorption prediction on tabular multi-factor data. Validated our choice to test both in Phase 4. Typical reported R ² values of 0.92–0.99 for ensemble models on adsorption data support our Phase 4 target of R ² (residuals) = 0.70.

Research gap identified	Review identifies a critical gap: “ensemble ML models are not combined with physics-based chemistry models for hybrid prediction.” The reviewed studies use ML as a black-box replacement for isotherms, not as a complement. This gap is exactly what our Langmuir + ML hybrid approach addresses.
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[32] (2022)

A Review of the Application of Machine Learning in Water Quality Evaluation *Eco-Environment & Health*, 1(2), 107–116 DOI: 10.1016/j.eehl.2022.06.001 <https://pmc.ncbi.nlm.nih.gov/articles/PMC9140000/> FREE (PMC Open Access)

What this paper is about	2022 comprehensive review of ML applications in water quality — monitoring, simulation, evaluation, and optimisation. Covers neural networks, decision trees, RF, SVM, and ensemble methods applied to pH prediction, contaminant detection, and treatment efficiency. Confirms that ML models outperform traditional regression approaches for multi-factor water quality problems (R^2 typically 0.88–0.97 vs 0.70–0.82).
What we used	Benchmarks for expected ML performance on water treatment data. The typical ML R^2 range of 0.88–0.97 supports our target for Phase 4. Confirmed that pH is the most frequently identified dominant predictor across water quality ML studies — consistent with our Phase 3 finding of pH as #1 residual driver.
Research gap identified	ML models reviewed are all pure data-driven approaches without physics integration. The review notes: “interpretability and alignment with known chemistry remain major challenges.” Physics-informed hybrid models are identified as the next frontier — precisely what our project implements.

[33] (2025)

Machine Learning vs Langmuir: A Multioutput XGBoost Regressor Better Captures Adsorption Dynamics *Soil Systems (MDPI)*, 5(4), 55 DOI: 10.3390/soilsystems5040055 <https://www.mdpi.com/2673-7655/5/4/55> FREE (MDPI Open Access)

What this paper is about	2025 study directly comparing Langmuir isotherm vs multioutput XGBoost for predicting soil phosphorus adsorption across 147 soils and 10,389 samples. XGBoost outperformed Langmuir in capturing the effect of Olsen P (predicted 12.6% drop vs Langmuir's underestimation) and sand content (19.2% drop). SHAP analysis confirmed the dominant predictors. Demonstrates that ML captures nonlinear multi-factor interactions the Langmuir model misses.
What we used	Most directly relevant 2025 paper to our project. Validates our finding that Langmuir alone ($R^2 = 0.8158$) misses systematic patterns that ML can learn. Confirms XGBoost as the recommended model for residual correction. The Langmuir-vs-XGBoost comparison paradigm is identical to our hybrid model concept applied to a different adsorption system (phosphorus instead of fluoride).
Research gap identified	The paper uses XGBoost as a direct Langmuir replacement rather than as a correction on top of the Langmuir baseline. Our project's hybrid approach (Langmuir + ML correction) is more novel and physically interpretable — we preserve the chemistry while correcting its systematic errors.

[34] (2025)

Application of Machine Learning Models in Optimizing Wastewater Treatment Processes: A Review *Applied Sciences*, 15(15), 8360 DOI: 10.3390/app15158360 <https://www.mdpi.com/2076-3417/15/15/8360> FREE (MDPI Open Access)

What this paper is about	2025 PRISMA-guided systematic review of ML and deep learning applications for wastewater treatment process optimisation. Covers ANN, RF, XGBoost, LSTM, CNN applied to biological treatment, coagulation, adsorption, and advanced oxidation. Key findings: XGBoost consistently achieves lowest error rates for process prediction; hybrid models (physics + ML) outperform pure ML; data scarcity is the main limitation.
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What we used	Confirmed XGBoost as the recommended model for adsorption process prediction (lowest MAPE, highest R^2 across reviewed studies). Validated the hybrid physics + ML framework as the emerging best practice, superseding pure ML approaches. The 80/20 train/test split used in 90% of reviewed studies validates our split choice.
Research gap identified	Review states: “integration of mechanistic models such as Langmuir isotherms with ML remains underexplored, with fewer than 5% of reviewed studies implementing hybrid approaches.” Of those few hybrid studies, none specifically target fluoride adsorption on bio-based carbon. Our project addresses this dual gap.

[35] (2024)

LHS Efficacy in Supervised Machine Learning — Conditioned and Progressive Approaches *Applied Numerical Mathematics*, 208, 256–270 **DOI:** 10.1016/j.apnum.2023.10.013 <https://www.sciencedirect.com/science/article/pii/S0168927423003240>

What this paper is about	2024 comparative study of standard LHS, Conditioned LHS (cLHS), and Progressive LHS (PLHS) for supervised ML training. Demonstrates that LHS-based training data consistently produces better-generalising ML models than random sampling of equal size. LHS training reduces required sample size by 30–40% for the same model accuracy.
What we used	Updated justification for using LHS (N=500) as our experimental design in Phase 1. The paper confirms that LHS-sampled training data improves ML generalisation — our 500-point LHS design in Phase 1 directly improves Phase 4 ML model quality compared to what random sampling of 500 points would achieve.
Research gap identified	The paper tests LHS benefits for classification ML tasks. Its application to regression-based hybrid models for adsorption prediction (our specific use case) is not studied. Our project provides a practical demonstration of LHS + hybrid ML for environmental engineering applications.

Summary Table — All 35 Papers

#	Author	Year	Topic	Key Contribution to Project	Open Access
1	Langmuir	1918	Isotherm theory	Core model equation	
2	Foo & Hameed	2010	Model comparison	Langmuir selection rationale	
3	Freundlich	1906	Freundlich isotherm	Rejected alternative	
4	Ho & McKay	1999	PSO kinetics	Kinetic term in simulation	
5	Lagergren	1898	PFO kinetics	PFO comparison; rejected	
6	Talat	2018	Coconut husk column	qmax, KL, design ranges	
7	Said & Machunda	2014	Coconut shell batch	pH, time validation	
8	Mondal	2015	Al-coconut fibre	Hardness 12% limit	
9	Bhomick	2019	KOH biomass carbon	Ea = 20 kJ/mol range	
10	Bhatnagar	2011	Fluoride review	pH bell curve, ion limits	
11	Sivasankar	2013	Carbonaceous mineral	pH mechanism, bell width	
12	Habuda-Stanić	2014	Ion effects review	All ion penalty limits	
13	John	2018	Anion competition	Cl ⁻ mechanism validated	
14	Newcombe	1997	NOM on carbon	NOM 15% fouling limit	
15	Mourabet	2012	Thermodynamics	Enaics 20,000 J/mol	
16	WHO	2017	Drinking water standard	C range, project context	
17	Fawell	2006	Fluoride in water	Project motivation	

#	Author	Year	Topic	Key Contribution to Project	Open Access
18	Guo	2018	ML for fluoride	RF precedent, pH dominance	
19	Breiman	2001	Random Forest	RF algorithm basis	
20	Chen & Guestrin	2016	XGBoost	Phase 4 ML algorithm	
21	Li & Sansalone	2024	Physics-informed ML	Hybrid model justification	
22	McKay	1979	LHS design	DoE method	
23	Pedregosa	2011	Scikit-learn	All ML computations	
24	(2024)	2024	Adsorption parameters	All factor ranges validated	
25	Mohapatra	2009	Fluoride review	Technology selection	
26	(2024)	2024	Global fluoride review	2024 context, novelty confirmed	
27	(2024)	2024	Agro-waste adsorbents	Coconut husk parameters updated	
28	Dubey	2026	Biosorbent bibliometrics	Research gap confirmed 2026	
29	(2024)	2024	Modified AC fluoride	pH, time, temp range confirmed	
30	Wang	2023	MgO-AC fibre	Ion competition 2023 confirmation	
31	(2022)	2022	Ensemble ML adsorption	RF/XGBoost as best methods	
32	(2022)	2022	ML water quality	pH dominant predictor confirmed	
33	(2025)	2025	ML vs Langmuir XG-Boost	XGBoost outperforms Langmuir	

#	Author	Year	Topic	Key Contribution to Project	Open Access
34	(2025)	2025	ML wastew- ater review	XGBoost recommended, hybrid needed	
35	(2024)	2024	LHS for ML training	LHS improves ML generalisation	

Open access: 22 of 35 papers (63%)

What the Recent Papers (2022–2026) Confirm

The 10 new papers from 2022–2026 collectively validate the novelty and importance of this project:

- 1. The research gap is still open (as of 2026):** Papers [26], [27], [28] all explicitly state in their 2024–2026 publications that no hybrid physics-ML model exists for coconut husk or bio-based carbon fluoride adsorption.
- 2. XGBoost is the right Phase 4 choice:** Papers [31], [33], [34] confirm XGBoost as the leading method for adsorption process ML, outperforming standalone Langmuir and traditional regression.
- 3. pH is the dominant predictor (confirmed 2022-2024):** Paper [32] confirms pH as the most frequently identified dominant predictor in water quality ML studies, directly supporting Phase 3 finding.
- 4. Hybrid models are the emerging best practice:** Papers [21], [34] confirm that physics + ML hybrid models outperform pure ML approaches and are identified as the next frontier.
- 5. LHS is validated for ML training (2024):** Paper [35] confirms LHS-sampled training data improves ML generalisation vs random sampling — directly validating our Phase 1 DoE choice.

Compiled: May 2026 / Project: Hybrid Chemical-ML Modelling of Fluoride Adsorption on Coconut Husk Activated Carbon Total references: 35 / Foundational: 25 / Recent (2022–2026): 10 / Open access: 22 (63%)